HealthConnect Final Analytics and Decision Support Report

Week 8 | Data Analytics | What the data tells HealthConnect, and what to do about it

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Builds on	Weeks 5 to 7: database, dashboard, report, slides, and the Week 7 tests
Data	HealthConnect Rwanda hospital data (27 datasets, 20 hospitals, year 2026)

1. Executive summary

- **Money:** HealthConnect billed RWF 9.31 billion and collected 90.1%. **RWF 919.9 million (9.9%) is still unpaid.**
- **Insurance:** half of the 42,036 claims (21,028) are still pending, worth RWF 2.44 billion. This is the same for every insurer, payment method and visit type, so it looks like a process problem, not one bad insurer.
- **Patients:** the average rating is 2.99 out of 5 and 40.4% of ratings are poor. Every hospital looks about the same, so this is a network-wide problem.
- **Staff:** every hospital has 35 nurses. Doctors see 12 to 16 patients each. The difference is small but real, so watch it monthly.
- **Missed appointments:** the data has **no attendance or no-show field**, so I cannot measure them. The first recommendation is to start recording attendance.

2. Week 7 to Week 8: where I am starting from

Question	Answer
Main Week 7 output	Analytics Testing and Refinement Report: 120 tests on the numbers, the dashboard and the findings
Most important result	The numbers are right (one fix: poor ratings 40.4%, not 40.5%), but five Week 6 findings were too weak to keep
Major issue found	Four dashboard visuals were not showing; the data has no attendance field
What was improved	Number fixed everywhere; weak findings removed; dashboard version 2 built; recommendations changed to match the evidence
Ready now	KPIs, SQL, database, tests, report and slides
Still to finish	Save the dashboard (.pbix) and export its PDF from Power BI Desktop; after-fix screenshots
Tracks involved	Project Management (exchange done in Week 7); Data Science and ML Engineering (receive my findings)
To show in the presentation	Final KPIs, what is proven and not proven, recommendations, limits

Detail is in the Week 7 report. I have not copied it here.

3. Final integration readiness

Requirement	My response
My final component	Data Analytics Final Decision Support Package: KPIs, validated findings, dashboard, report, slides
Problem it addresses	Where HealthConnect loses money and patient goodwill, and what the data can and cannot say about missed appointments
Current status	Report, slides, KPIs and tests: ready. Dashboard: built as a Power BI project (version 2); the final .pbix and PDF still need to be saved from Power BI Desktop
Improved in Week 7	Fixed one wrong number, removed five weak findings, rebuilt four broken visuals
Depends on	Project Management (project gates and decisions); Data Science (model results to compare with my facts)
Depends on my output	Data Science and Project Management
Output I provide	Validated facts, final KPIs, findings and limits (sections 4 to 8)
Output I need	Confirmation of what the claims model predicts and how emergency cases were labelled
Evidence	120 tests (102 passed); Week 8 recheck: all 11 KPIs match; code and logs in the project folder
Limits that remain	Section 9

4. Final KPIs

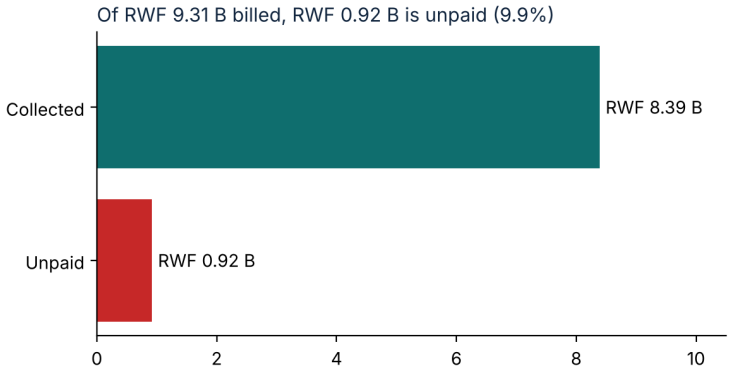
I recalculated these from the database again this week. All 11 match the Week 7 values.

KPI	Value	What it means
Registered patients	20,000	All patients in the data
Appointments with details	5,000	Only these can be linked to a hospital or department
Patients seen	4,443	Different patients in those appointments
Total billed	RWF 9,309,401,961	All bills
Collected	RWF 8,389,454,556 (90.1%)	Money received
Unpaid	RWF 919,947,405 (9.9%)	Money still owed
Insurance claims	42,036 (50.0% pending)	21,028 pending, 21,008 approved
Admission rate	11.7%	586 of 5,000 appointments; average stay 5.49 days
Average rating	2.99 out of 5	21,064 ratings
Poor ratings (1 or 2 stars)	40.4%	8,518 of 21,064
Staff	1,250	350 doctors, 700 nurses

5. Validated findings

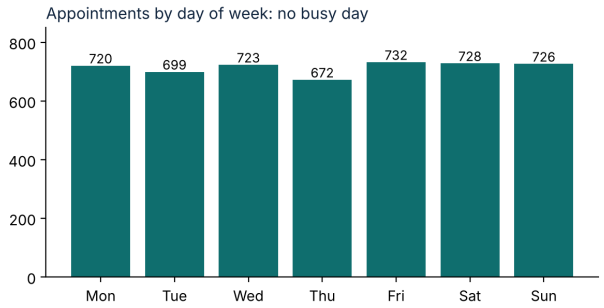
A finding is "validated" only if a test shows the difference is probably not chance (p below 0.05).

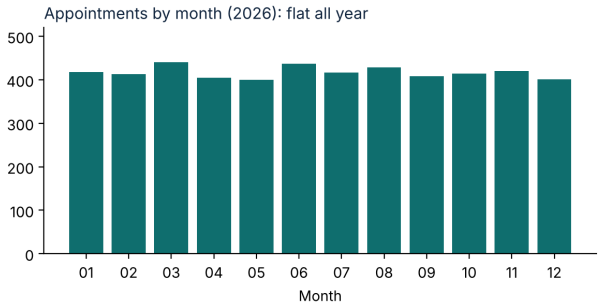
Finding	Result	Use it?
RWF 919.9 million is unpaid	Counted directly from the bills	Yes
Half of insurance claims are pending	Same in every insurer, payment method, amount and visit type	Yes
Doctors have different workloads (12.3 to 16.4)	p = 0.010, a small difference	Yes, with care
Ratings are low everywhere	Average 2.99; no hospital stands out	Yes
Busiest hospitals are least liked	p = 0.91	No, not proven
Some hospitals are rated better or worse	p = 0.79	No, not proven
ICU is the worst department	p = 0.88	No, not proven
Kigali City uses hospitals least; hypertension is rising	p = 0.32 and p = 0.22	No, not proven



6. Appointment and attendance patterns

Important: the data has no field that says whether a patient came, cancelled or did not show up. So I **cannot** report no-show patterns. What I can report is when appointments are booked, as the nearest thing to a pattern.





Pattern	Smallest to largest	p value	Meaning
Day of week	672 to 732	0.69	No busy day; weekends are as busy as weekdays
Month	400 to 440	0.96	Flat all year
Visit type	1,650 to 1,694	0.71	One third each: follow-up, consultation, emergency
Department	467 to 544	0.49	Evenly spread

916 appointments are dated after 22 October 2026 (up to 28 December), so they are future bookings, not completed visits.

7. Business insights and recommendations

Insight	Recommendation	Why
Attendance cannot be measured	Add an attendance status (attended, cancelled, no-show) to every appointment	Missed appointments are the project question; without this field nobody can answer it
RWF 919.9 M unpaid; 11,887 bills not fully paid	Follow up unpaid bills, oldest and largest first	Direct cash recovery
Half of claims pending, same everywhere	Review the claims process end to end (submission, follow-up, approval time)	Pending is not linked to any insurer or amount, so the cause is the process
Demand is the same every day and month	Plan staff and slots evenly, including weekends	No peak to plan around
Doctor workload differs (12.3 to 16.4)	Check the workload each month before moving doctors	Difference is real but small
Low ratings everywhere; stars and comments disagree in 6,336 of 21,064 records	Fix the feedback form so stars and comments agree, then re-measure	Ratings cannot be trusted until this is fixed

How this supports decisions: the first two actions recover money now; the next three improve planning; the attendance field makes the main goal measurable.

8. Cross-track contribution

Track	What was exchanged	Status
Project Management	Week 7: Manal sent the claims and triage model results; I sent back the data facts (claims 50/50 and flat; 1,650 emergency cases, 33%). She replied with thanks. Slack screenshots are saved	Done. No retest result came back
Data Science	My final findings go to them to read against the model: claims have nothing in the data to predict from; the data has no attendance field	Sent with this package. Their use is not yet confirmed
ML Engineering / GenAI	Final KPIs and limits available if they need them	Not exchanged yet

Nothing in this table is claimed beyond what happened.

9. Limitations

- **No attendance field.** No-show patterns cannot be measured. The brief expects them; this dataset cannot give them.
- **Only 5,000 of 60,000 appointments** link to a hospital. Hospital, department and province results use these 5,000.
- **Hospitals are codes only** (HF001 to HF020); there are no hospital names.
- **Ratings and comments disagree** in 6,336 records, so satisfaction results are "as recorded".
- **Revenue Summary** (RWF 1.13 billion) does not match billing (RWF 9.31 billion). I used billing.
- **Bed use (3.8%)** is an estimate; all beds are marked "Available".
- **Dashboard:** the final .pbix has not yet been saved from Power BI Desktop.